Xiang Li

lixiang.stu@yangtzeu.edu.cn · github.com/stardustlil · stardustlil.github.io

Education

The Hong Kong University of Science and Technology (Guangzhou) From Sep 2027
MPhil (incoming)

Yangtze University, Jingzhou, China Sep 2022 – Jun 2026
B.Eng. in Software Engineering
Relevant coursework: Discrete Mathematics, Probability & Statistics, Machine Learning, Algorithm Design & Analysis, Linear Algebra

Research

GeoRouteNet: Geometry-Enhanced Neural Solver for Euclidean TSP Nov 2025 – Jun 2026
Undergraduate thesis · arXiv:2606.22776

- Started from reproduction and ablation studies of NAR4TSP, which motivated a geometry-aware architecture for Euclidean TSP.
- Proposed a non-autoregressive neural solver that injects geometric inductive biases into GNN node, edge, and attention representations.
- Designed a multi-candidate self-comparative reinforcement learning strategy for training.
- Awarded University-Level Outstanding Graduation Thesis.

Competitive Programming & Awards

University Competitive Programming Team Oct 2024 – Jun 2026

- Solved 2000+ algorithmic problems across Codeforces, AtCoder, and LeetCode, focusing on dynamic programming, graph algorithms, and combinatorics.

National First Prize (rank 219/2520), Mati Cup Programming Contest Aug 2025
National Second Prize, RAICOM Robot Developer Competition (CAIP track) Aug 2025
National Second Prize, Lanqiao Cup Programming Contest (C/C++) Jun 2025

Selected Projects

poetryGPT: Decoder-only Transformer for Chinese Poetry Generation May 2025 – Jun 2025
PyTorch, CUDA · Independent project

- Built a 7-layer decoder-only Transformer (~14M parameters) in PyTorch for classical Chinese poetry generation.
- Developed a character-level preprocessing pipeline over 2.2M Chinese characters and implemented temperature, top-k, and nucleus sampling for autoregressive decoding.

Skills & Languages

- **Programming:** Python, C/C++, Java, SQL
- **Research & ML:** Literature reading and paper reproduction; implementing deep learning models in PyTorch; familiar with reinforcement learning concepts
- **Algorithms:** Data structures and algorithms, with extensive practice in dynamic programming, graph algorithms, and combinatorics
- **Languages:** Chinese (native), English (CET-6)